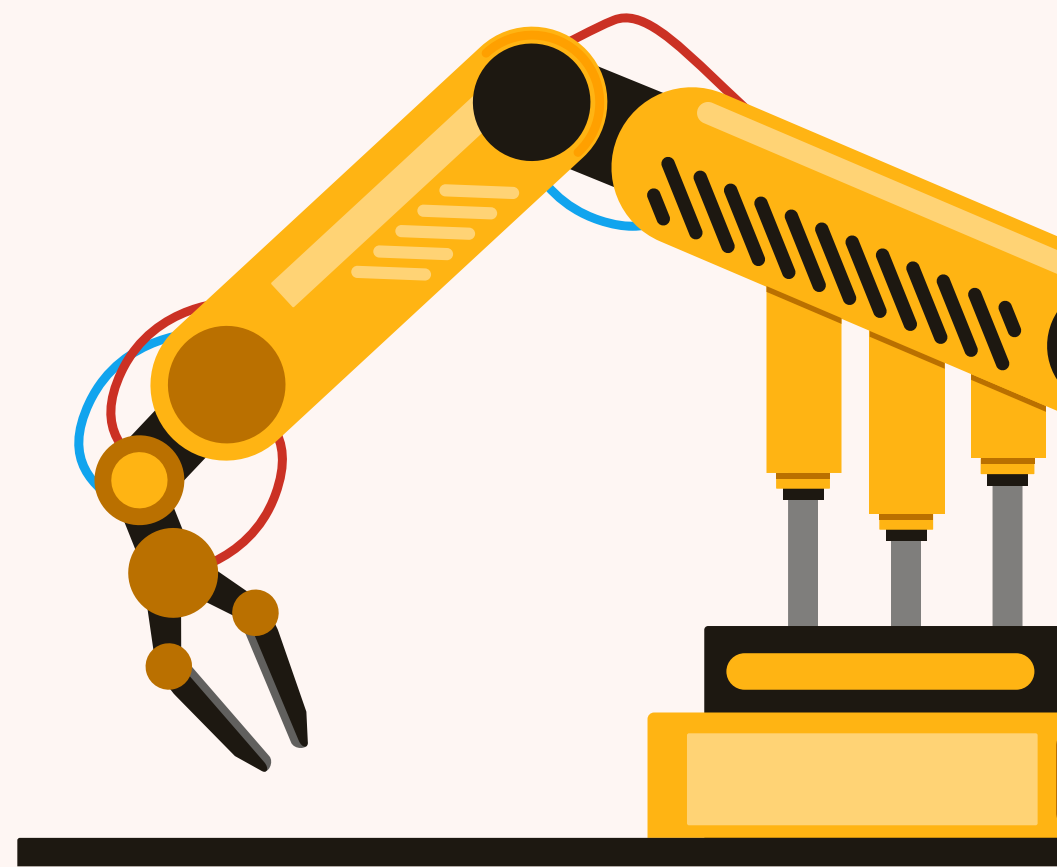
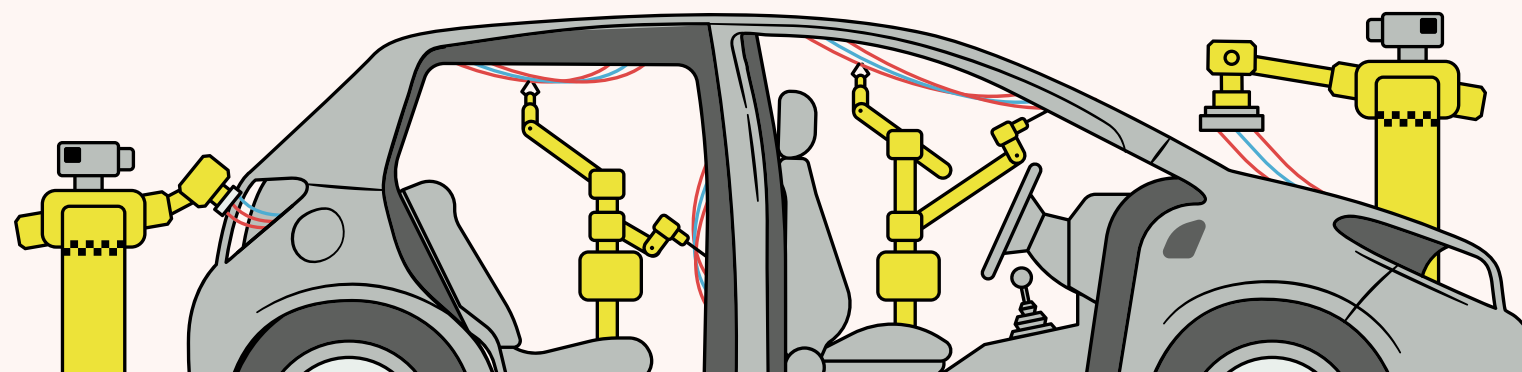
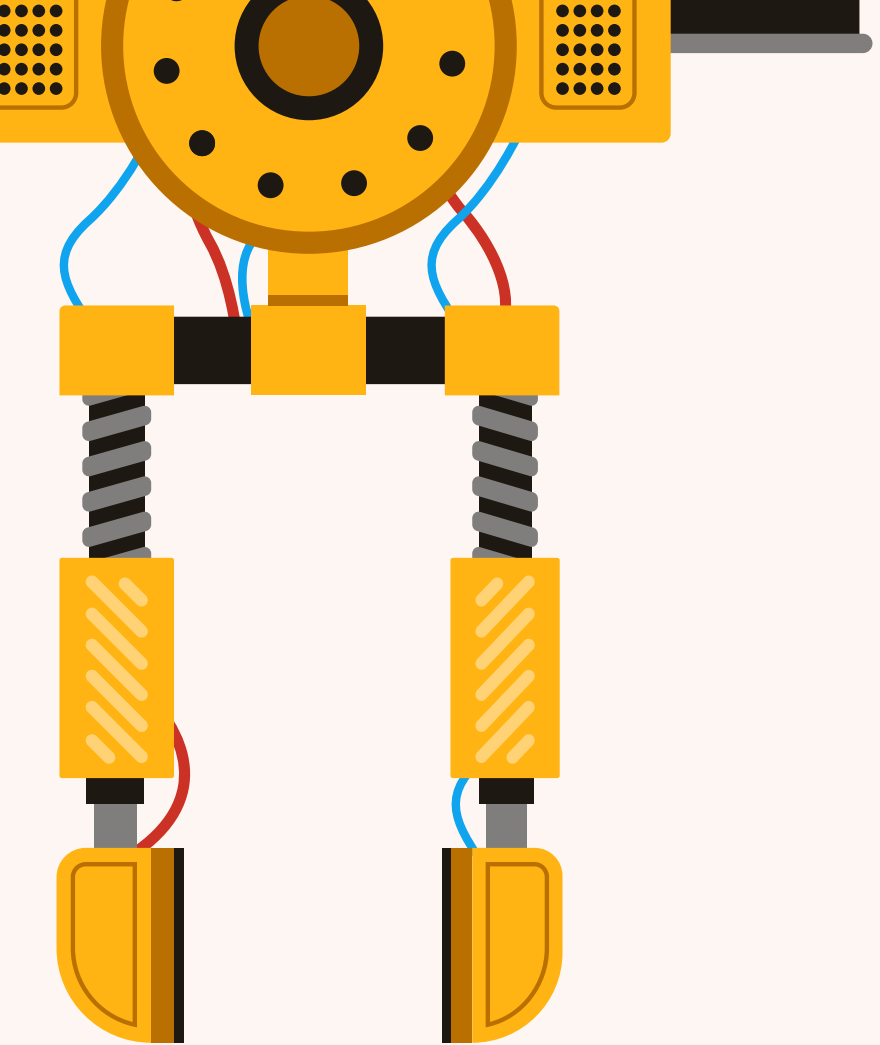
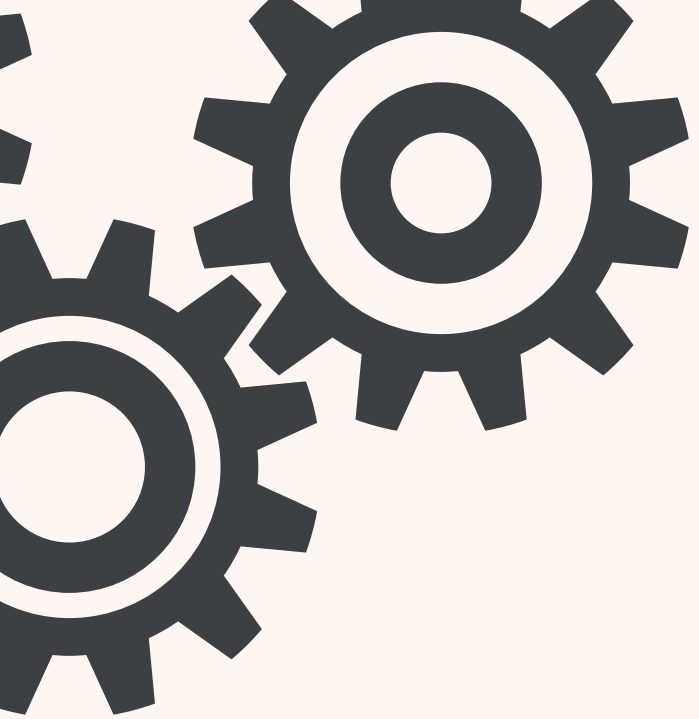


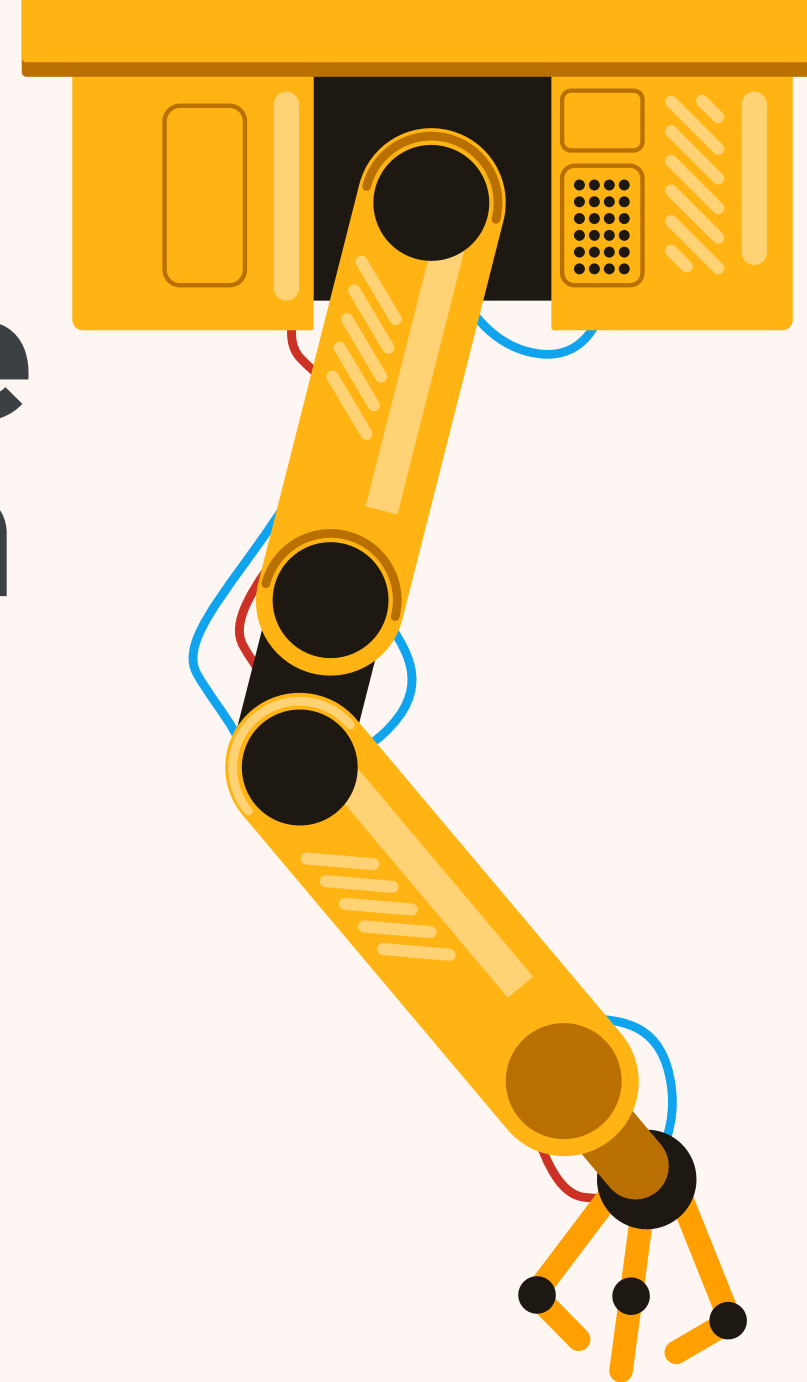
# My Report

## Presentation



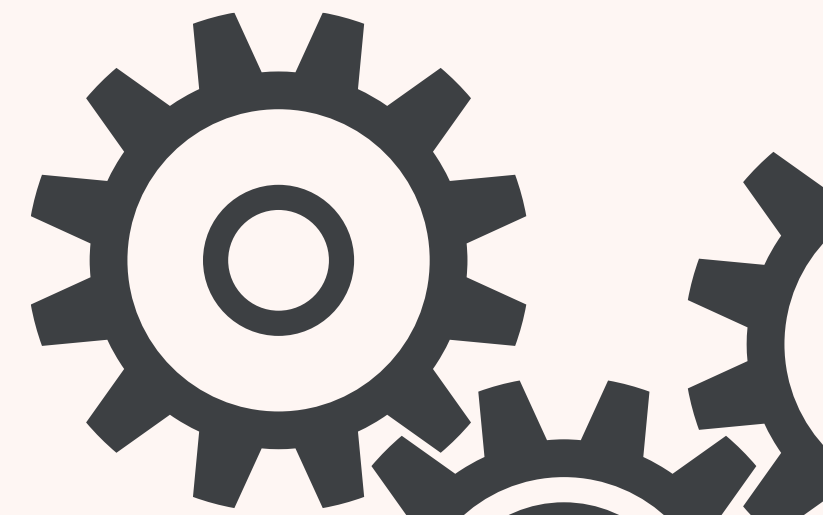
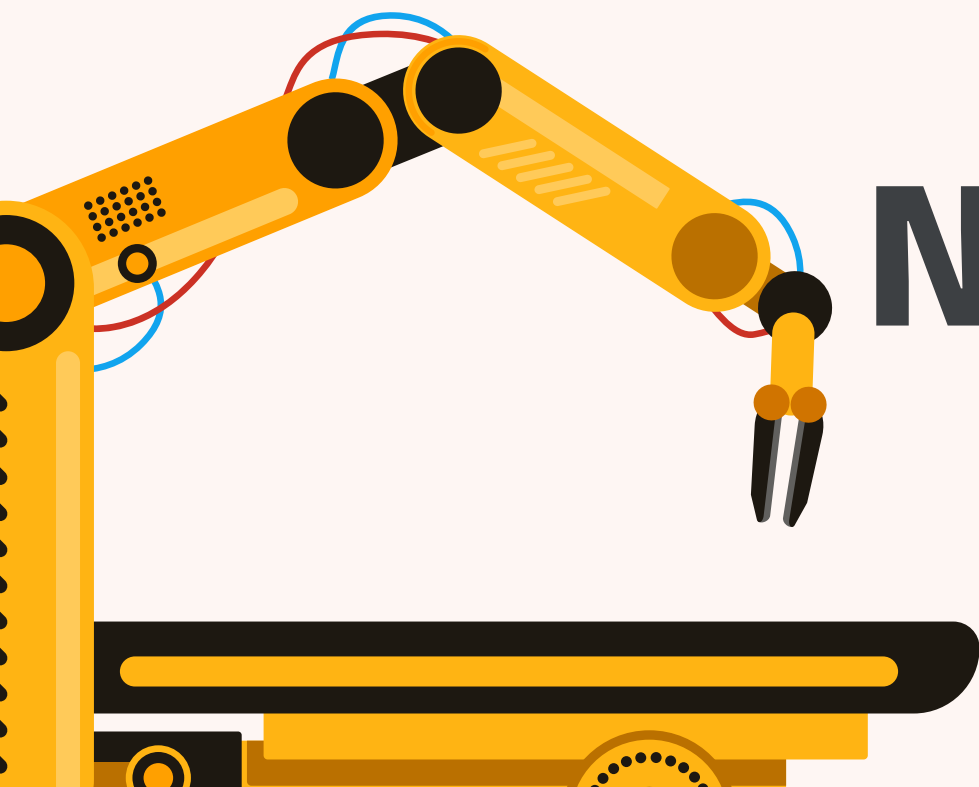


# **A Detailed Comparative Study of the Monongah and Courrières Mining Disasters**



**Presented**

**Nathaniel G. Pedronio**



# Introduction

Mining has historically been one of the most hazardous engineering industries due to:

- Confined Underground Environments
- Presence of combustible gases and coal dust
- Dependence on Mechanical Ventilation System

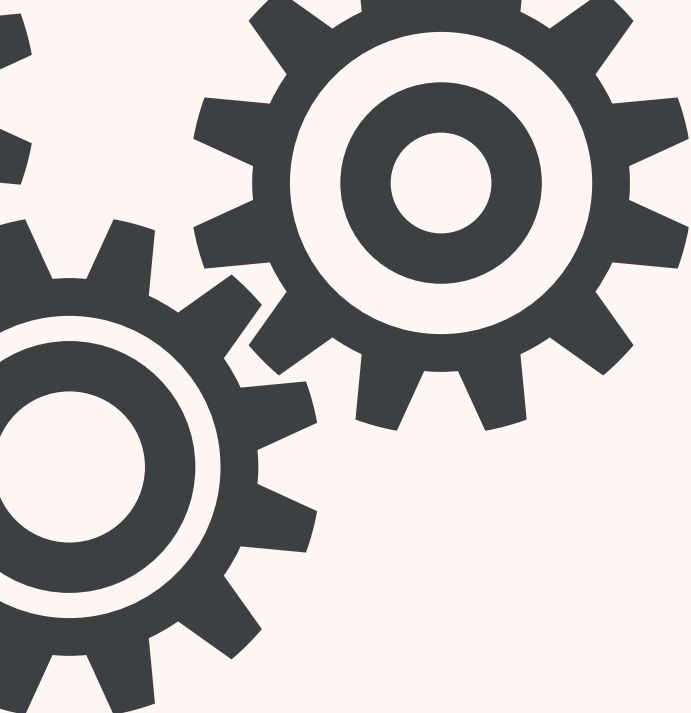


This presentation compares two of the deadliest mining disasters:

- **Monongah Mining Disaster (USA, 1907)**
- **Courrières Mining Disaster (France, 1906)**

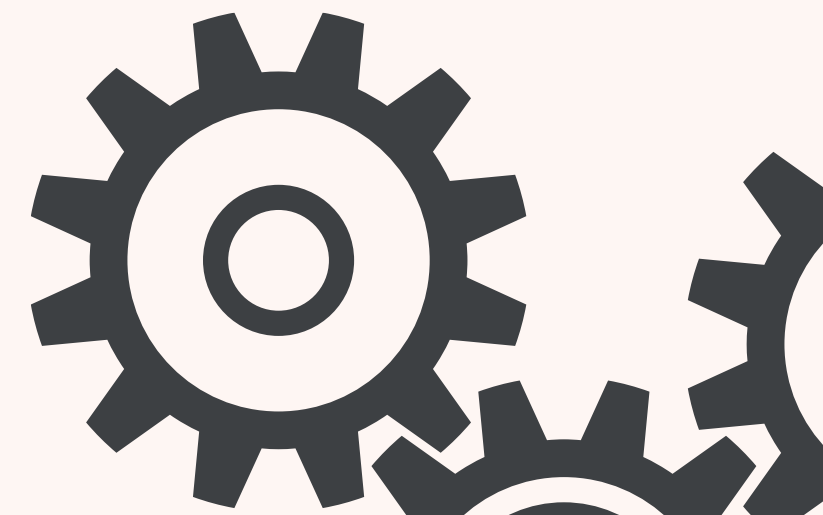
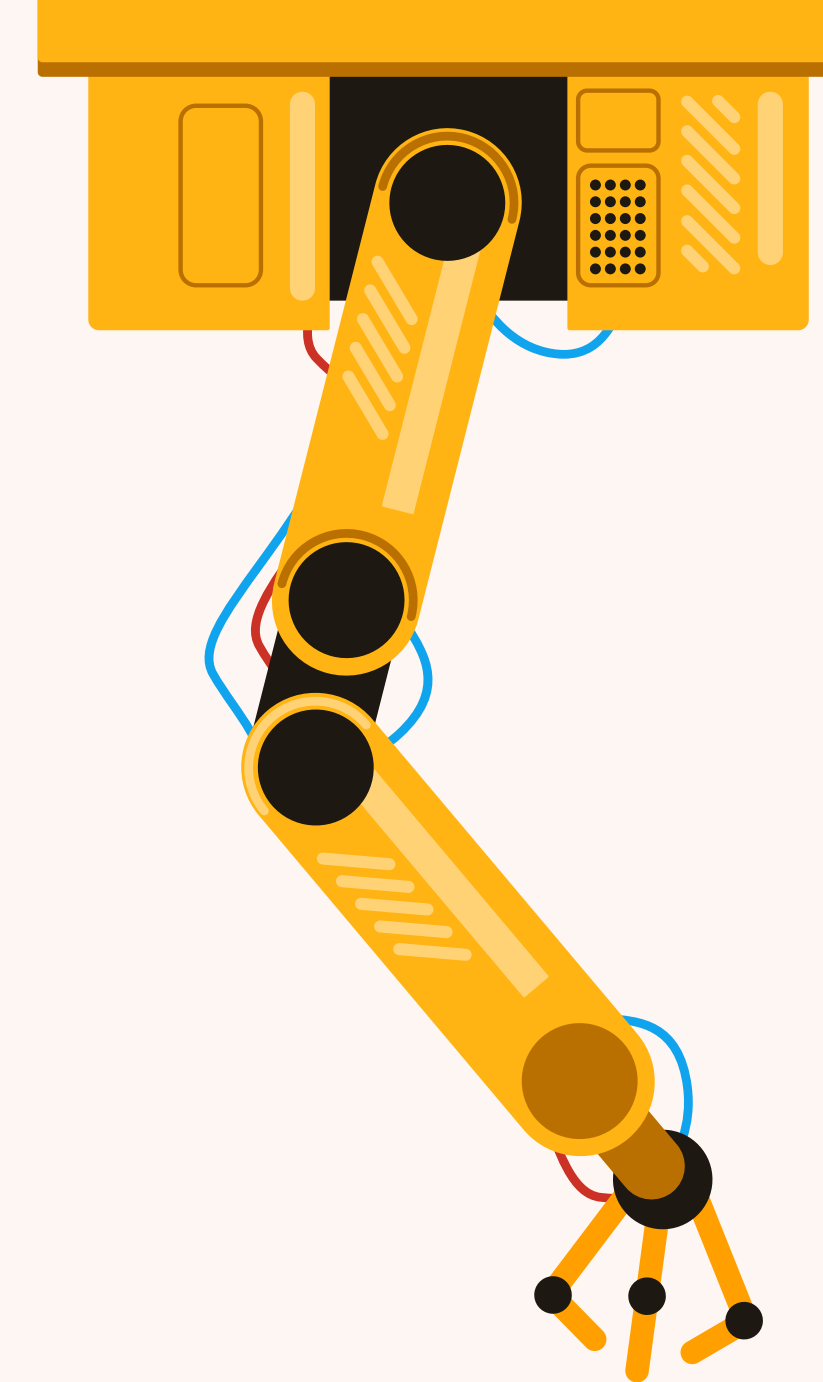
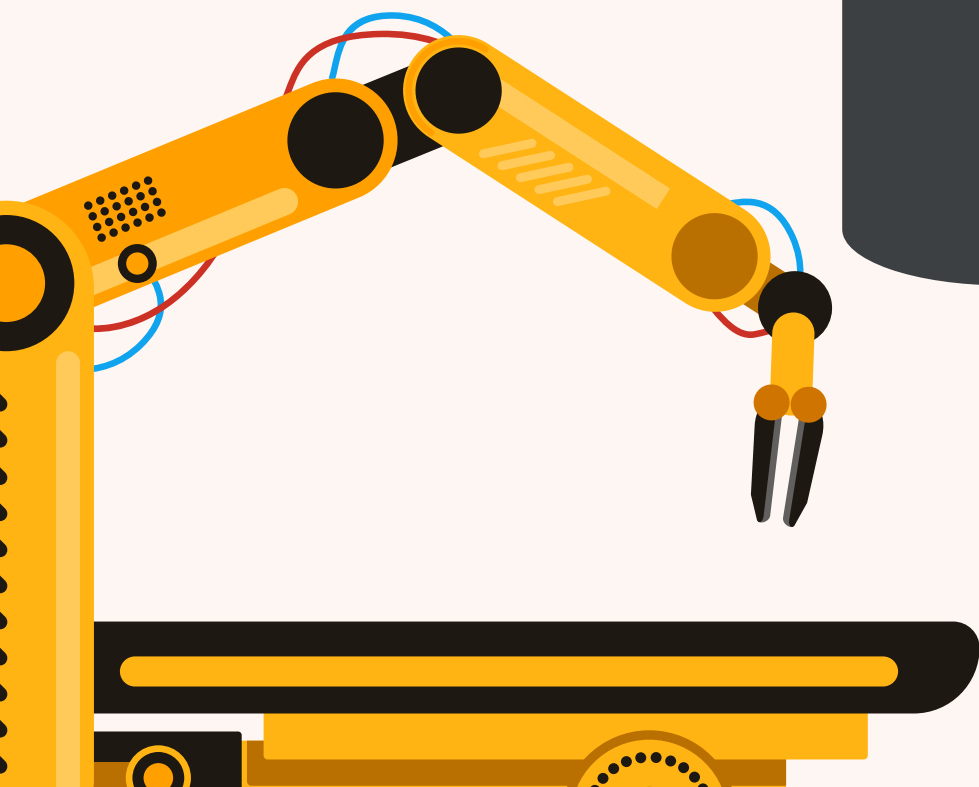
**Focus:**

Engineering failures, mechanical system limitations, and safety lessons.



# Objective of the Study

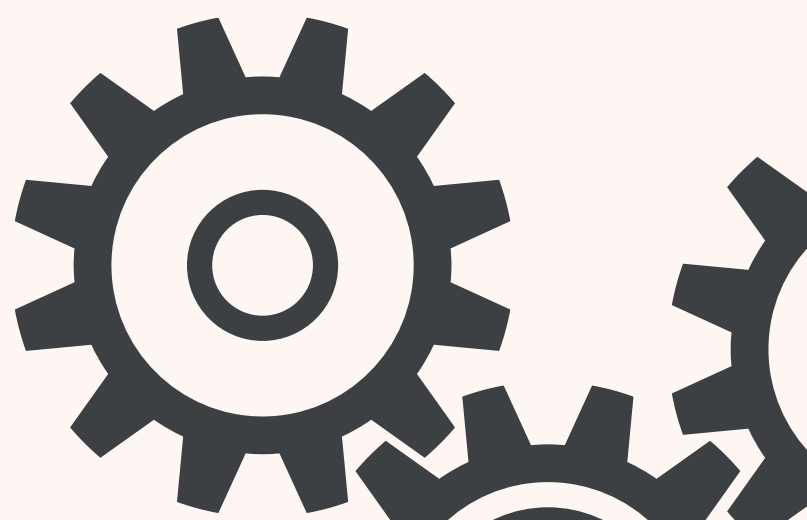
- To analyze the technical causes of both disasters
- To identify mechanical engineering system failures
- To compare similarities and differences
- To extract engineering lessons applicable to modern safety design

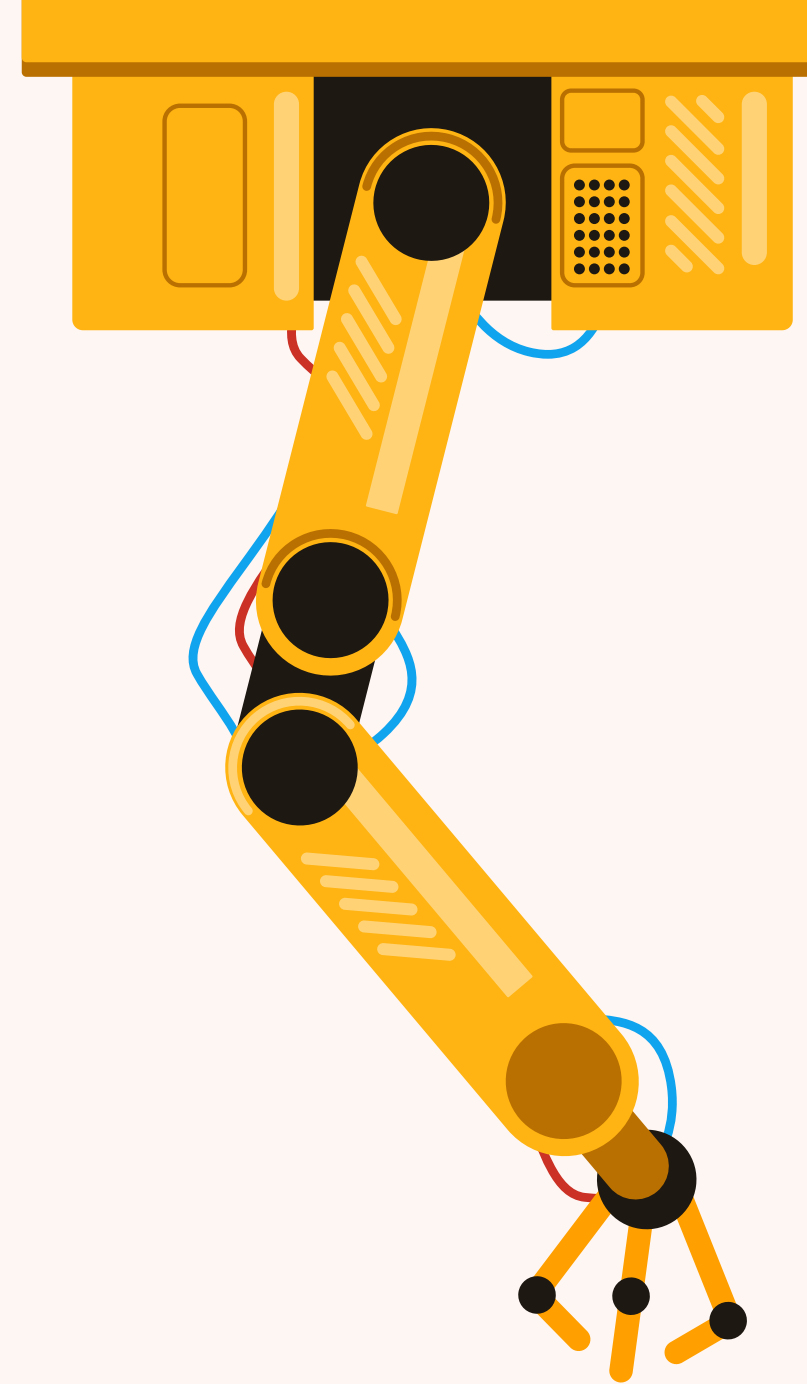
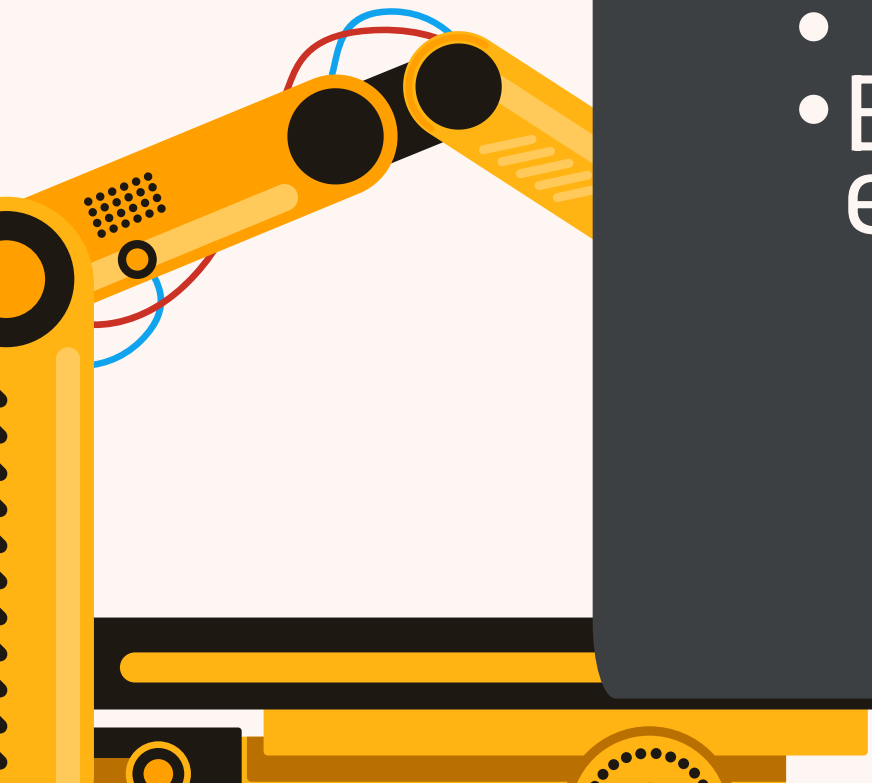


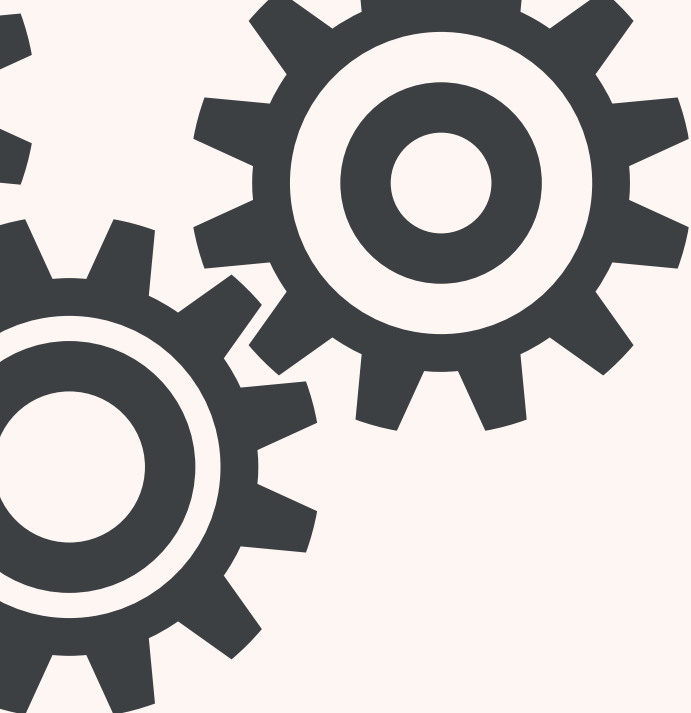




# Background: Mining Hazards

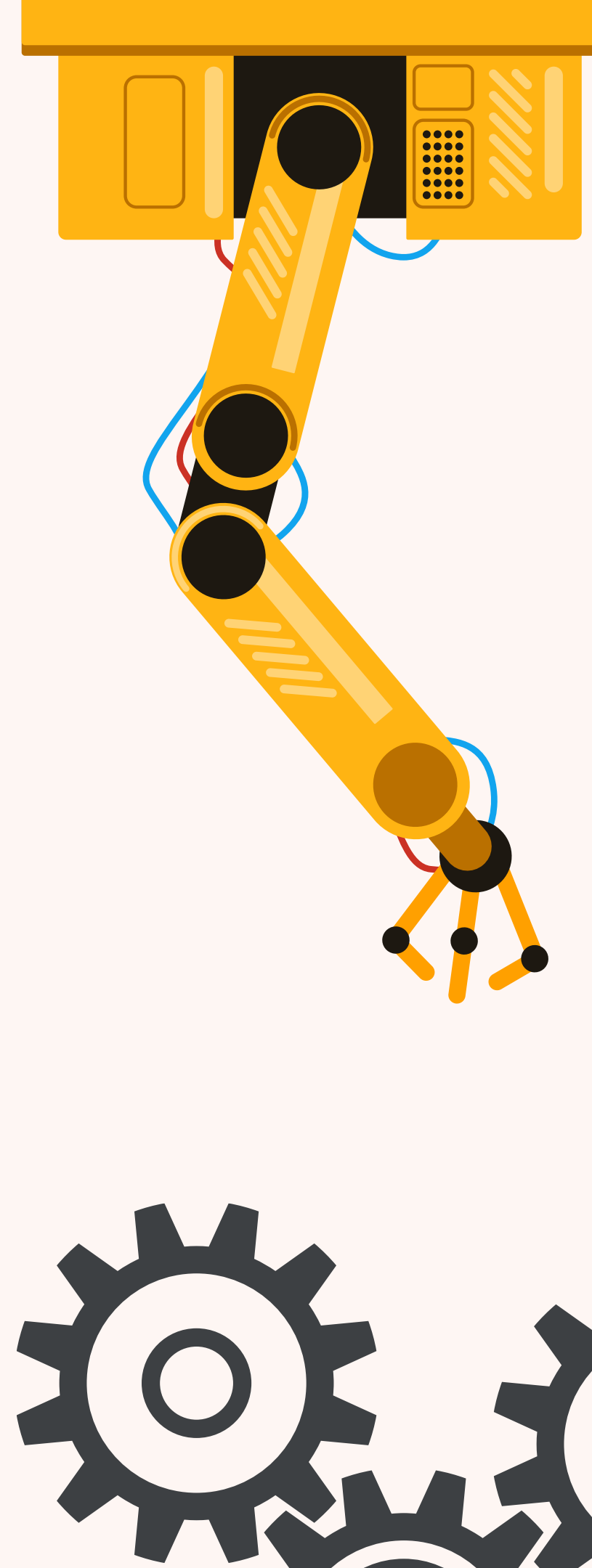
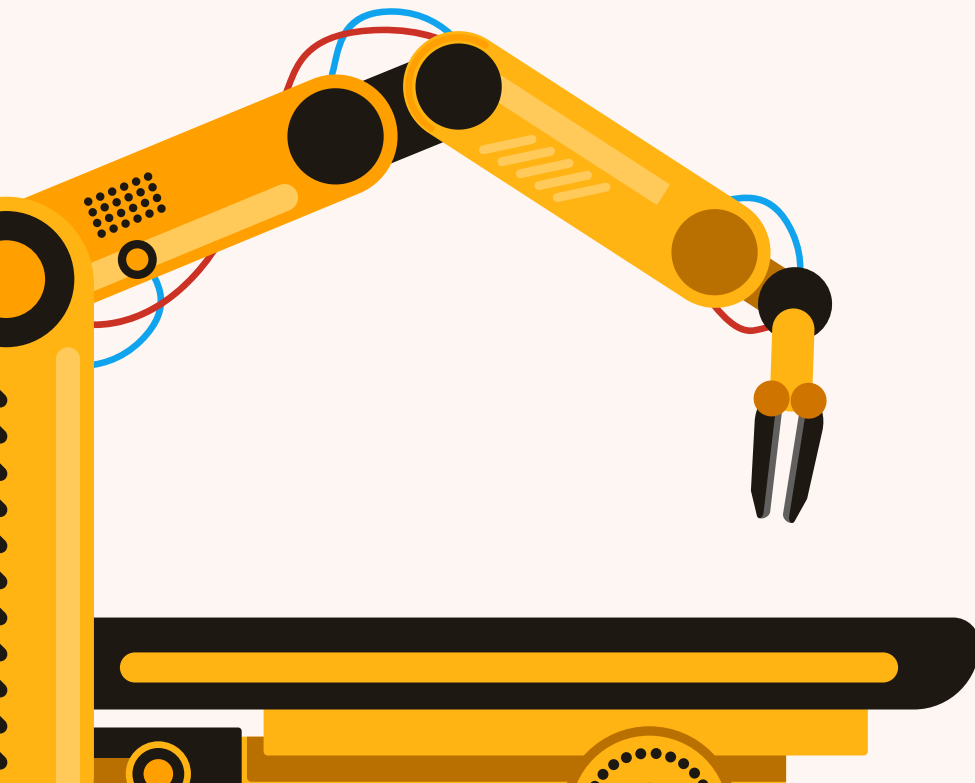
- Major hazards in underground coal mining:
  - Methane gas accumulation
  - Coal dust suspension
  - Inadequate ventilation
  - Mechanical equipment sparks
  - Poor emergency escape design
  - Engineering responsibility: Control energy, airflow, and ignition sources.
- 





# Monongah Mining Disaster Overview

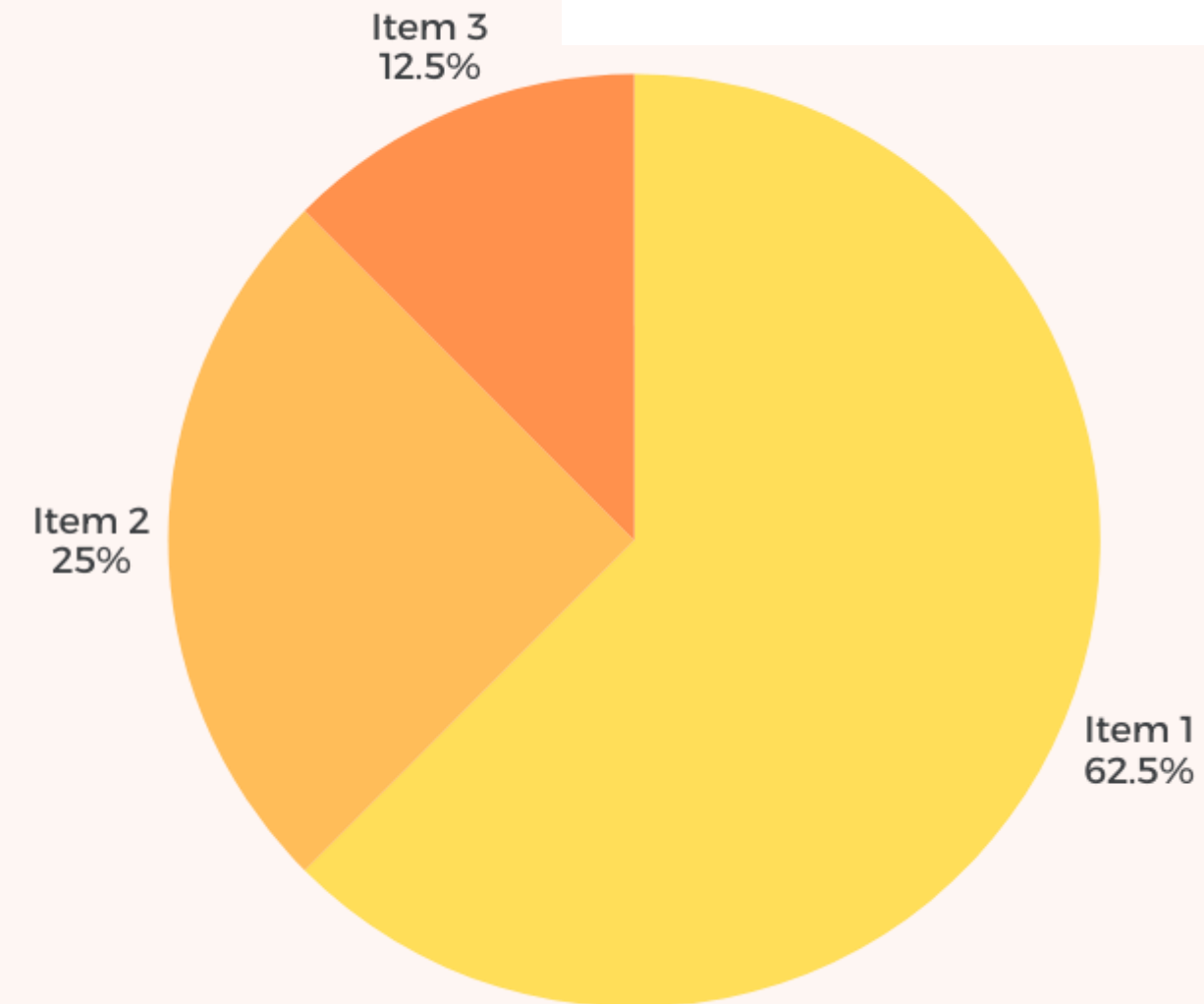
Date: December 6, 1907  
Location: West Virginia, USA  
Mine Type: Bituminous coal mine  
Fatalities: At least 362 miners  
Significance: Deadliest mining disaster in U.S. history



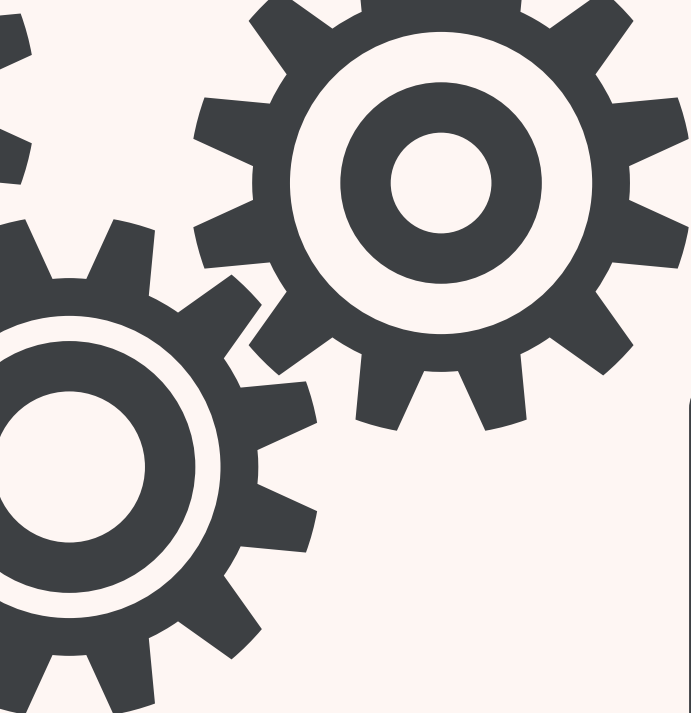
# Result

## Rescue and Recovery Efforts

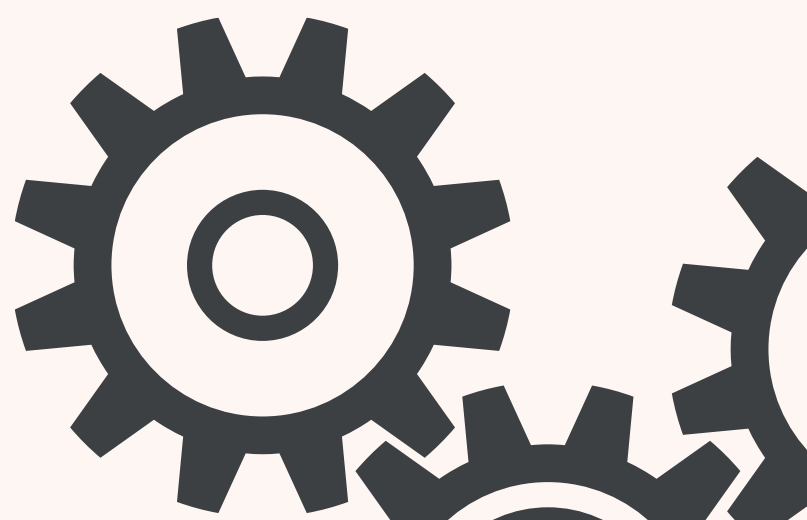
- Rescue teams faced:
  - Toxic gases
  - Collapsed tunnels
- Limited technology slowed rescue
- Survivors were extremely few
- Bodies were recovered for weeks

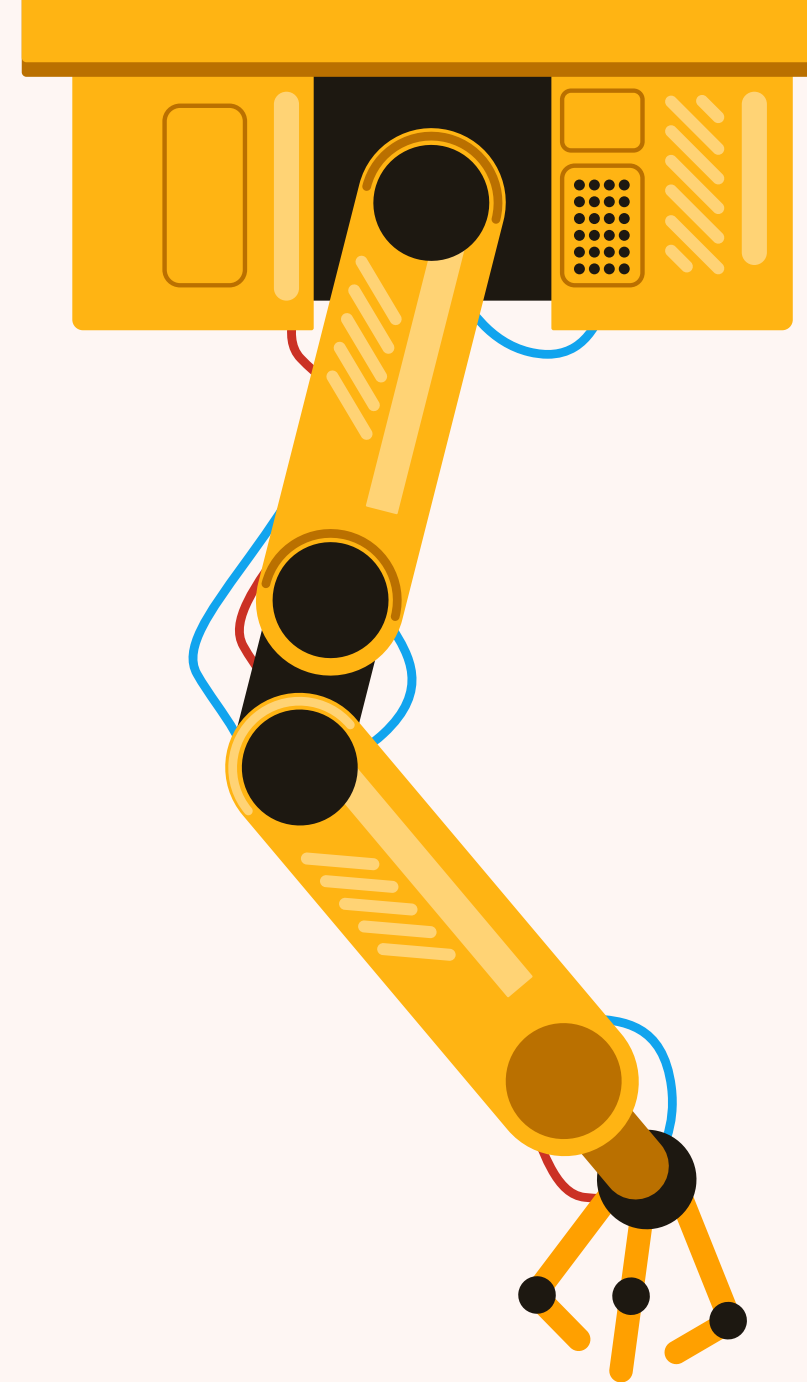
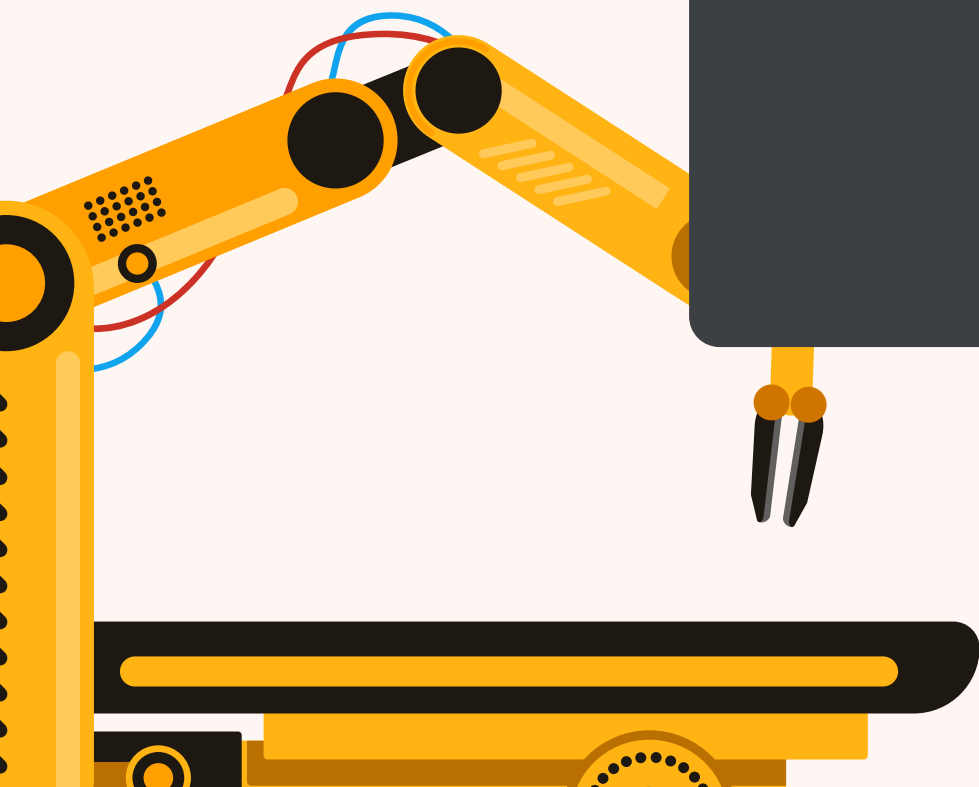


65.5 % Lack of Structural Analysis  
25 % Technology  
12.5 % Poor Decision Making

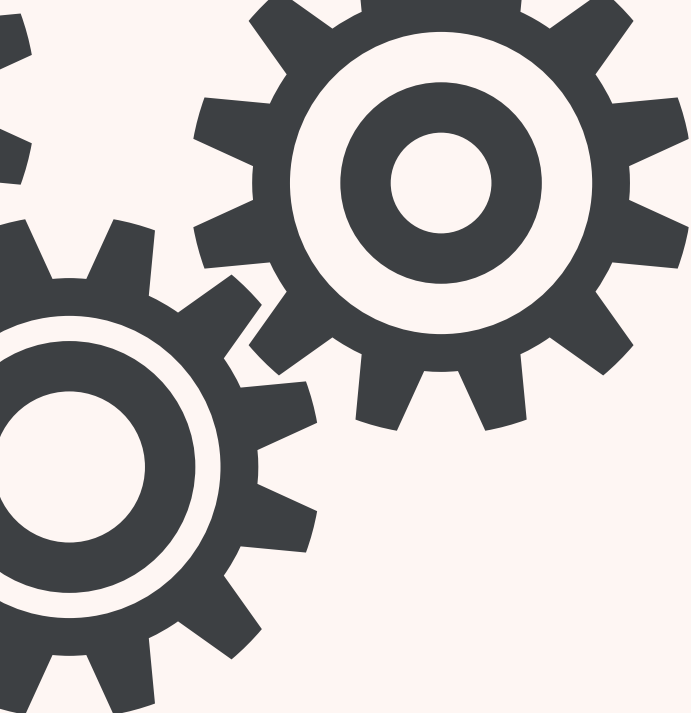


# Monongah Disaster: Technical Causes

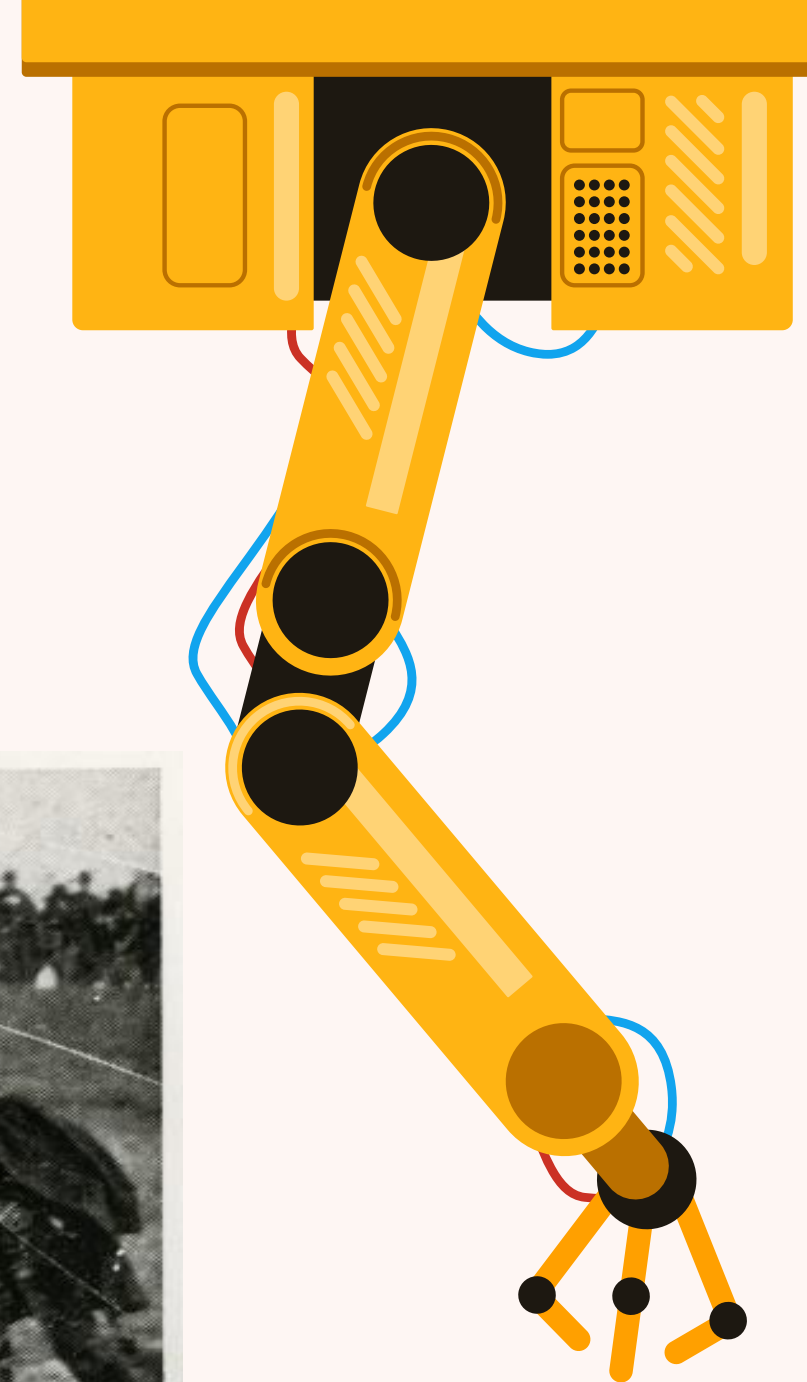
- Accumulation of methane gas ( $\text{CH}_4$ )
  - Presence of suspended coal dust
  - Ignition source likely from:
    - Open flame lamps
    - Electrical spark
  - Ventilation system failure
  - Result: Massive underground explosion
- 





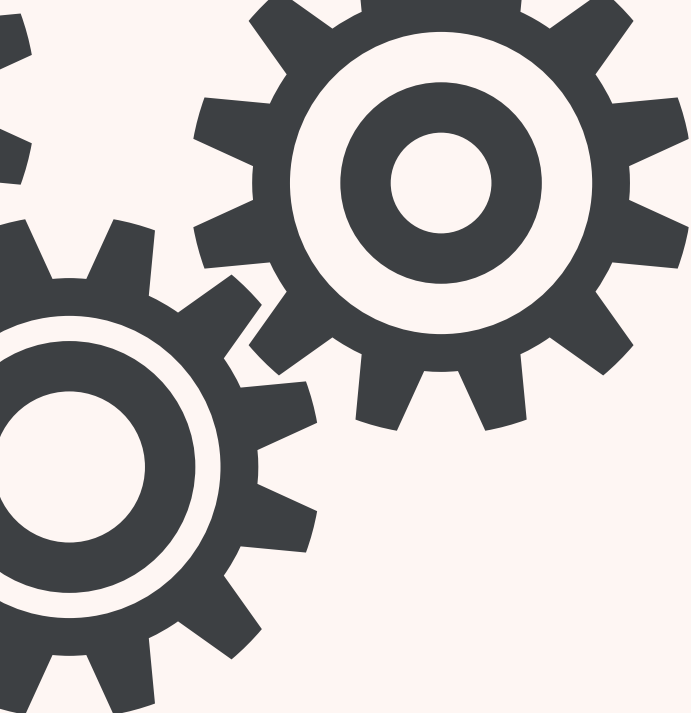


## Engineering and Safety Lessons

- Importance of ventilation design
  - Gas monitoring systems are critical
  - Risk assessment must be continuous
  - Worker safety should be prioritized over production
  - Engineering ethics matter
- 

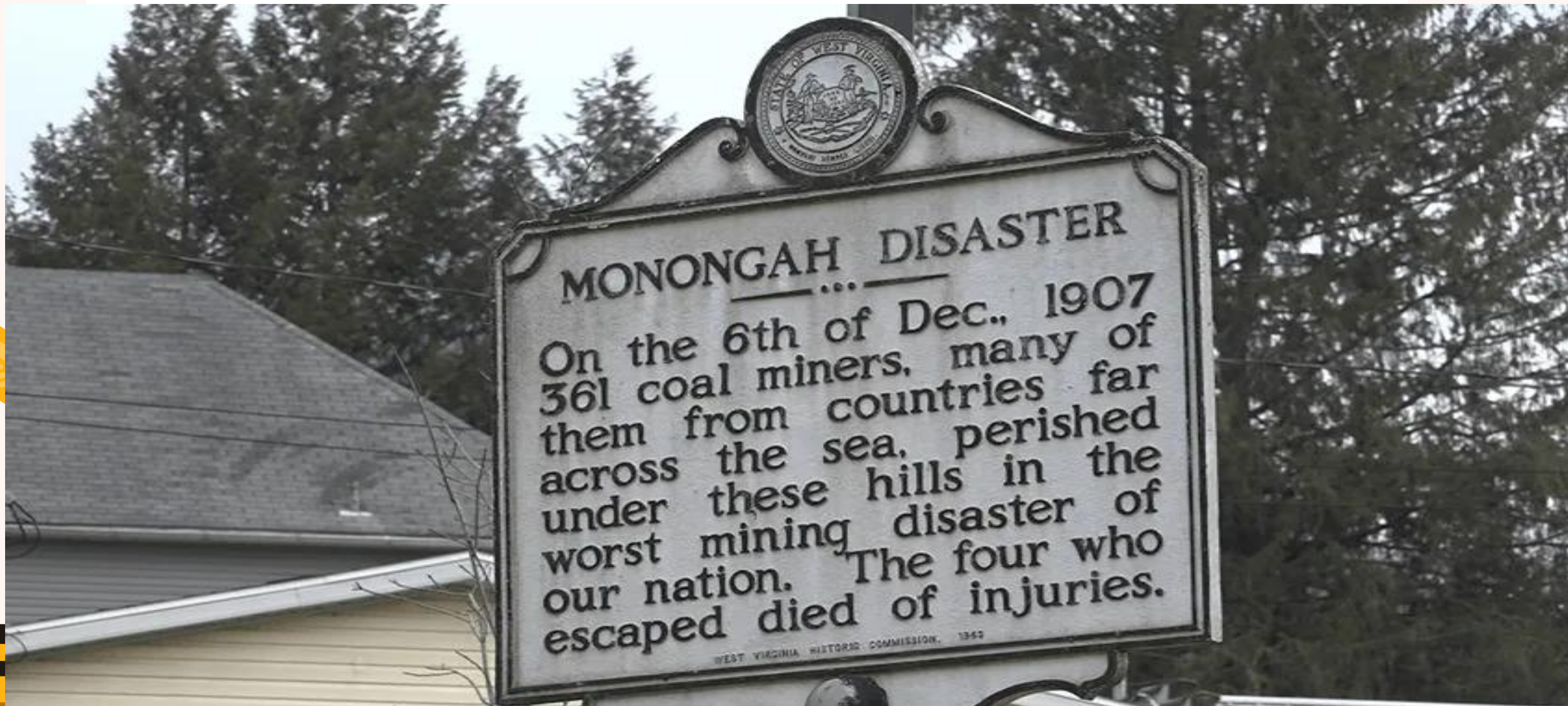






## Relevance to Engineering Students

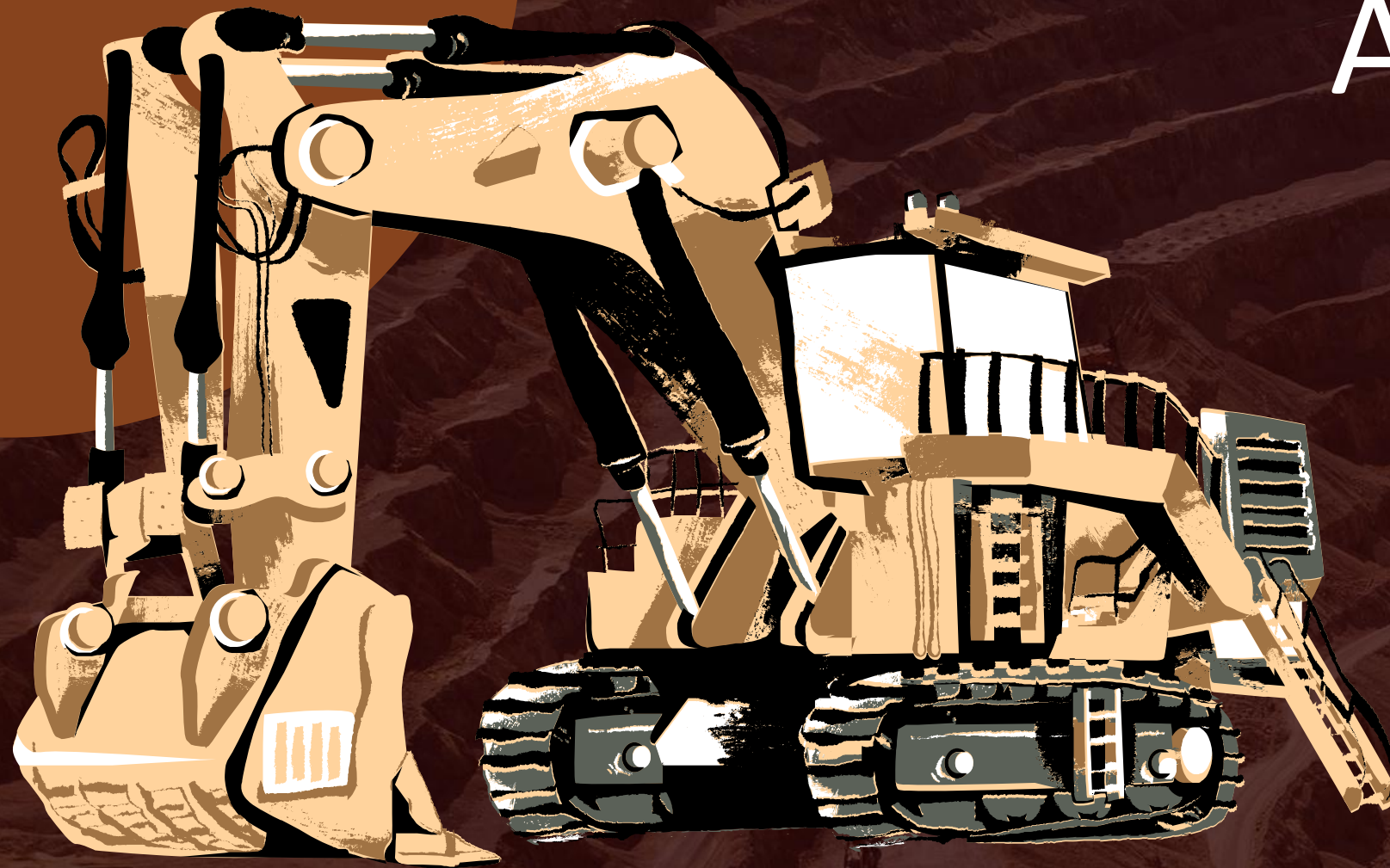
- Demonstrates consequences of poor design and management
- Highlights need for safety standards
- Encourages responsibility in engineering decisions
- Applies to:
  - Mechanical Engineering
  - Mining Engineering
  - Industrial Safety





Presentation Title

# Engineering Failure Analysis



The Courrières Mine Disaster  
(March 10, 1906)

Presented By: Nathaniel G. Pedronio

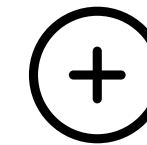




*To understand the disaster, we must look at the mine as a complex mechanical system involving fluid dynamics and thermal equilibrium.*

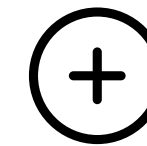


### **Ventilation Architecture**

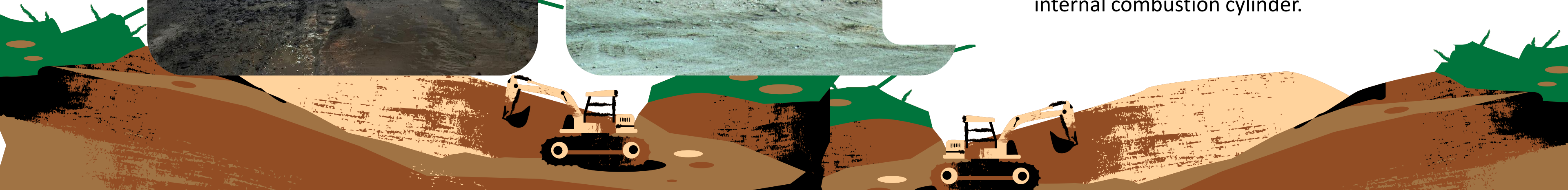


- A network of interconnected shafts designed to circulate fresh air and extract methane (firedamp).

### **The Fuel Source**



- High-volatile coal dust. In mechanical terms, the mine acted as a massive, uncontained internal combustion cylinder.





## Courrières

### Mining Disaster Overview

**Date:** March 10, 1906

**Location:** Pas-de-Calais, Northern France

**Mine Type:** Coal mine with interconnected tunnels

**Fatalities:** 1,099 miners

**Significance:** Deadliest mining disaster in Europe





## **Courrières Disaster: Technical Causes**

- Coal dust explosion (primary cause)
- Inadequate ventilation control
- Delayed response to early fire signs
- Long interconnected tunnels allowed pressure wave propagation

**Explosion traveled over several kilometers**





## Technology Needed

### Stone Dusting

- Adding inert pulverized limestone to "dilute" coal dust and stop combustion propagation.

### Self-contained breathing apparatus (SCBA)

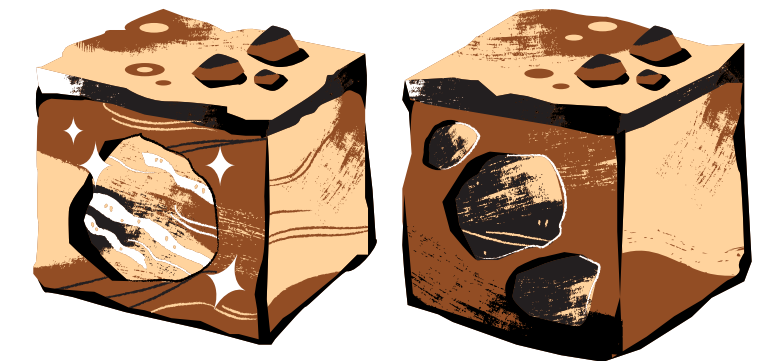
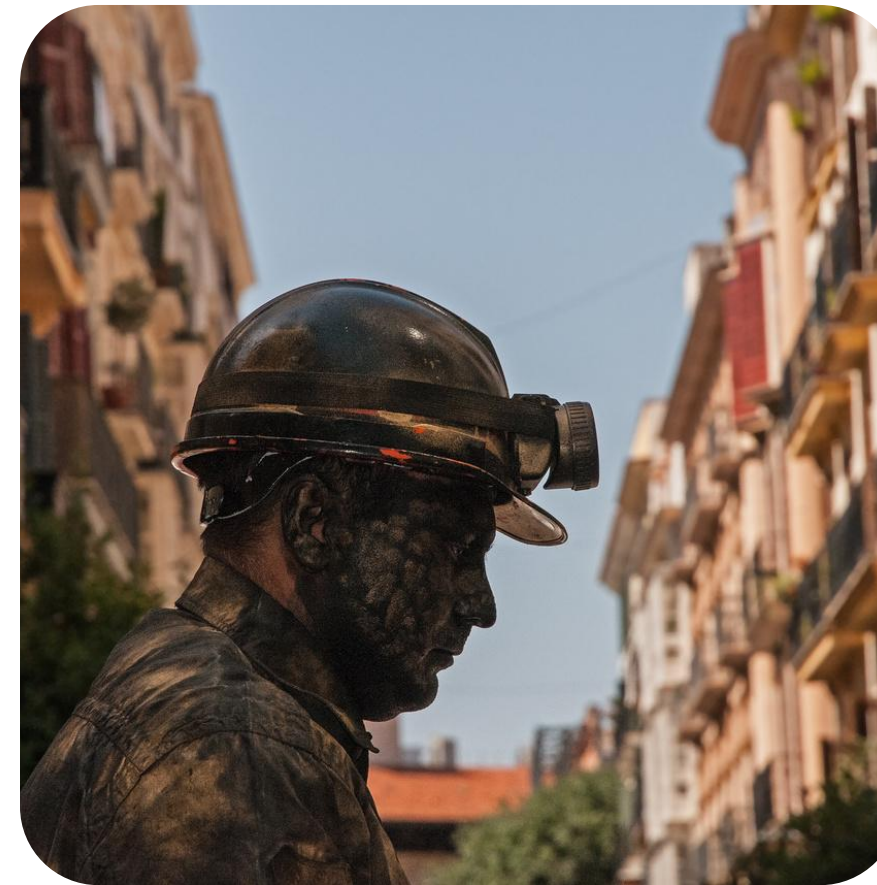
- Creating a portable closed-loop oxygen system for rescuers.

### Explosion-Proof Housing

- Designing electrical and mechanical equipment to contain internal sparks.

### Ventelatsison Monitoring

- Implementation of automated pressure and gas sensors to detect leaks early.





# Lessons for Modern Engineers



## **Redundancy:**

One fan failure should not collapse the entire life-support system.



## **Dust Control:**

Managing particulate matter is as critical as managing flammable gases.



## **Human Factors:**

Engineering for "survivability" means designing exit paths that remain viable even after a structural compromise.



# Ventilation System Comparison

| → Aspect           | Monongah                  | Courrières                 |
|--------------------|---------------------------|----------------------------|
| → Ventilation type | Natural / weak mechanical | Poorly designed mechanical |
| → Airflow control  | Minimal                   | Inconsistent               |
| → Gas monitoring   | None                      | None                       |
| → Dust suppression | Absent                    | Absent                     |





# Impact & Aftermath



**1**

Strikes and protests



**2**

Improvements in mine safety laws



**3**

Labor movement strengthened in  
France



Anong engineering improvements ang dapat ginawa para maiwasan ang parehong sakuna? 😊





**Ventilation** - the essential process of supplying fresh air and controlling airflow in underground mines to ensure a safe, breathable environment.

**Pressure control** - refers to the management, monitoring, and regulation of the intense forces exerted by surrounding rock and specialized equipment to ensure safe and efficient extraction. It involves managing two main types of pressure

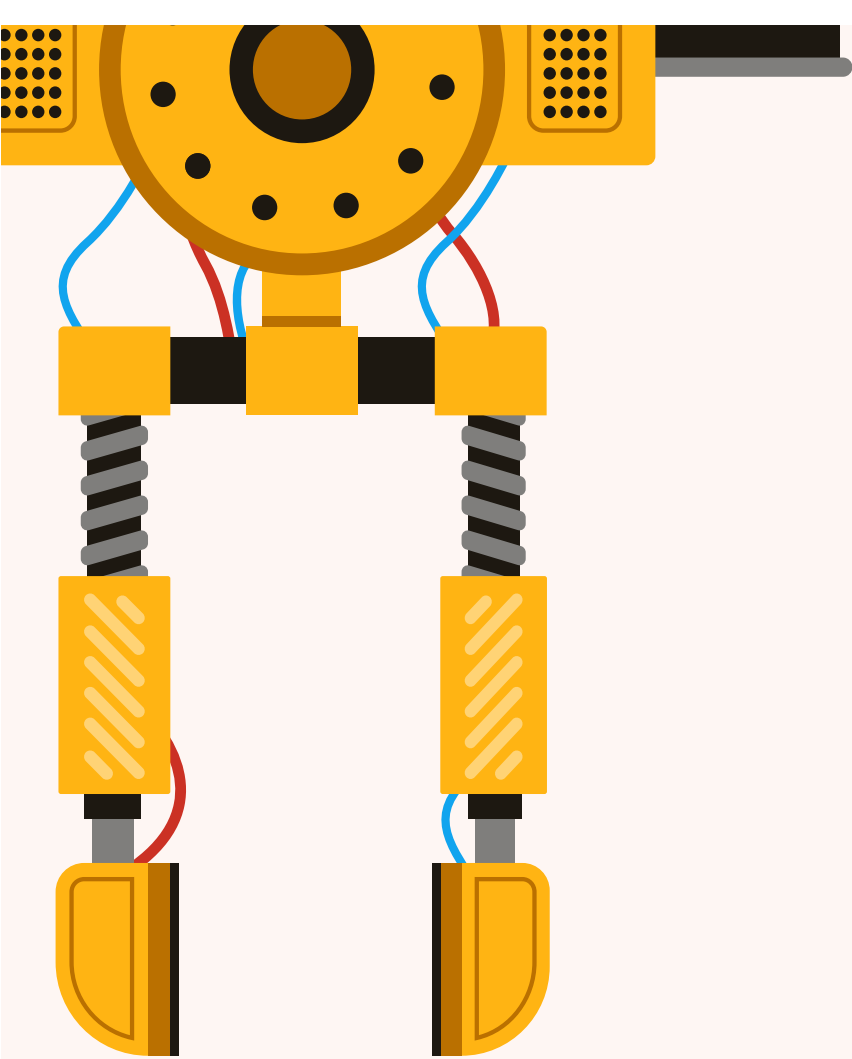
**System redundancy** - refers to the intentional inclusion of duplicate or backup components, systems, or pathways within mining infrastructure to ensure continuous operation, safety, and reliability, even if a primary component fails.



Parehong disaster ang  
magpapasabi sa isang ME  
student na “*Hindi lang pala  
exam ang pwedeng  
bumagsak—pati buong  
system kapag walang safety  
design!*”







# Thank You

